	OL	T	H	IN	K
1	Sunny	Hot	High	weak	No
2	Sunny	mild	High	strong	No
3	cloudy	Hot	High	weak	Yes
4	Rainy	mild	High	weak	Yes
5	Rainy	cool	Normal	weak	Yes.

ENTROPY :-

Yes = 3 No = 2

$$H(S) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$H(S) = +0.442 + 0.528$$

$$H(S) = 0.97$$

IG For OL

OL	Yes	No	Total	Entropy
Sunny	0	2	2	0.00
cloudy	1	0	1	0.00
Rainy	2	0	2	0.00

$$H(S|Sunny) = -\frac{0}{2}\log_2\left(\frac{0}{2}\right) - \frac{2}{2}\log_2\left(\frac{2}{2}\right)$$

$$H(S|cloudy) = 0 - 0$$

$$H(S|Rainy) = 0$$

Date

IG For T

T	Yes	No	total	Entropy
Hot	1	1	2	1
mild	1	1	2	1
cool	1	0	1	0

$H(S|Hot) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$

$H(S|mild) = 1$

$H(S|cool) = 0$

IG For H

H	Total	Yes	No	Entropy
High	4	2	2	1
Normal	1	1	0	0

$H(S|High) = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$

Normal = 1

$H(S|[illegible]) = 0.00$

IG For W

W	Total	Yes	No	ENT
weak	4	3	1	0.811
strong	1	0	1	0

Signature

No.

$H(S|weak) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\frac{1}{4}$

$= 0.811$

$H(S|strong) = 0.00$

IG calculations:-

$IG(OL) = 0.0865 - 0 = 0.0865$

$IG(T) = 0.0865 -$

HS calculations:-

$H(S|OL) = \frac{2}{5}(0) + \frac{1}{5}(0) + \frac{2}{5}(0)$

$= 0$

$H(S|T) = \frac{2}{5}(1) + \frac{2}{5}(1) + \frac{1}{5}(0)$

$= 0.8$

$H(S|H) = \frac{4}{5}(1) + \frac{1}{5}(0)$

$= 0.8$

$H(S|W) = \frac{4}{5}(0.811) + \frac{1}{5}(0) = 0.811$

Date

IG :-

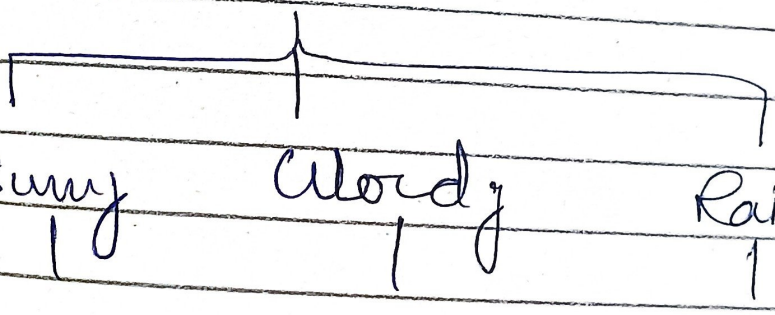
$$IG(Ol) = 0.97 - \frac{0.97}{0.97} \times 0 = 0.97$$

$$IG(T) = 0.97 - \frac{0.97}{0.97} \times 0.8 = 0.97 - 0.8 = 0.17$$

$$IG(H) = 0.97 - \frac{0.97}{0.97} \times 0.8 = 0.17$$

$$IG(W) = 0.97 - \frac{0.97}{0.97} \times 0.811 = 0.159$$

Ol is root node.



Signature

No.